

The Structural Folding Front

From Domain Application to Structural Runtime and General Fold/Unfold Intelligence

Unified ResearchGate One-Page Overview of SMSF, CSFR, and GFSFUI - v1.0.0

Key idea. The Structural Folding Front studies how concrete experience can be transformed into reusable structural memory and selectively activated or Unfolded under new goals and contexts. A useful Fold preserves enough structure for localization, search, comparison, validation, and future Unfolding; a useful Unfold reuses certified structure where possible and introduces only the necessary new structural difference. The three repositories form a coordinated progression from a demanding application, to reusable runtime mechanisms, to a general Fold/Unfold intelligence lifecycle.

Unified progression: Concrete Application -> Reusable Structural Runtime -> General Fold/Unfold Framework
Lifecycle: Experience -> Fold -> Structural Memory -> Localize -> Search -> Unfold -> Validate -> Grow -> Refold
Persistent safeguards: Explicit Leftover + Policy Governance

SMSF

Stock-Market Structural Folding

Role: Concrete Application / Proof by Construction

Core focus

- Historical market experience -> reusable structural memory
- XYM-style structural evidence and Pattern Differential Tree
- Localization, evidence composition, policy, and decision
- Preserve alternatives rather than collapse experience into one prediction
- Heavy Fold, Light Unfold

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GitHub: <https://github.com/sizhet/Stock-Market-Structural-Folding-SMSF>

CSFR

CCC Structural Folding Runtime

Role: Reusable Algorithm / Runtime / Infrastructure

Core focus

- Metric representation and structural clustering
- Pattern-to-Pattern, Pattern-to-CCC, and CCC-to-CCC metrics
- Cluster CCC as a structural possibility set rather than a centroid
- Claw-Dragon structural merge
- CCC DNA, reverse indexing, and Two-Phase Search
- Retrieve First, Verify Second

DOI: 10.5281/zenodo.22650524

GitHub: <https://github.com/sizhet/CCC-Structural-Folding-Runtime-CSFR>

GFSFUI

General Framework of Structural Folding and Unfolding Intelligence

Role: General Framework / Fold-Unfold Lifecycle

Core focus

- Represent -> Fold -> Localize -> Search -> Challenge -> Unfold -> Validate -> Grow -> Refold
- Counter-evidence and structural falsification
- Sparse-data candidate structure and continual structural growth
- CallingGraph localized Unfolding: Certified Core + Minimal Necessary Delta
- Explicit Leftover as a competence boundary
- Policy-governed, user-tunable structural runtime

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GitHub: <https://github.com/sizhet/General-Framework-of-Structural-Folding-and-Unfolding-Intelligence>

Reading map: Stock teaches Folding. Biomedical structural search illustrates Search, Counter-Evidence, and Growth. CallingGraph illustrates Unfolding, Generation, and Refolding. **Research principle:** Compute-by-Need, Growth-by-Need, Unfold-by-Need - global structural memory with a small active intelligence footprint.

Canonical research records: The GitHub/DOI repositories contain the complete version-controlled research materials. This one-page document is an overview and navigation artifact.